

ELECTIVE COURSE-3.3

URA735: AUGMENTED REALITY AND VIRTUAL REALITY				
	L	T	P	Cr
	2	2	0	3.0
Course Objective: To provide students with the knowledge and skills to design, develop, and implement immersive and engaging AR and VR experiences for a variety of applications.				
Syllabus				
Introduction to AR & VR: Overview, history, definition: Virtual Reality, Augmented Reality, Applications: Remote Robotic System, surgery, e-commerce, industrial design.				
Computer Graphics: Raster images, ray tracing, linear algebra, transformation matrices, surface shading, texture mapping, data structure for graphics.				
Vision system: Light and Optics, physiology of human vision, visual perception, visual rendering system.				
Body motion and forces: Accelerometer, gyrometers, magnetometers, complementary filter, Kalman filter, Measurement of velocities, accelerations, tracking position and orientation, tracking attached bodies.				
Sound and haptic Rendering Systems: Physiology of human hearing, audio perception, audio rendering: processing, modelling, auralization, haptic sensors, force feedback, tactile .				
Comfort and VR sickness: motion sickness, visually induced sickness, simulator sickness, measurement of various sickness				
Laboratory Work				
1. Explore the Unity editor, identify its essential features, and navigate in 3D space. 2. Create and manipulate game objects, manage scene, and manage projects in Unity 3. Development of a basic game project-1 4. Development of a basic game project-2 5. Basics of VR project and implementing locomotion and grabbing objects				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

6. Implement events and interactions in VR for immersive experience: audio, haptics, & UI 7. Explore ergonomics and optimization in VR: comfort, optimization, and lighting
Projects: Students will be submitting two projects on augmented reality for applications such as automotive industry, education, robotic system, etc.
Course Learning Objectives (CLO) The students will be able to: - define and explain the fundamental concepts of AR and VR, including their history, applications, and underlying technologies. - develop and implement simple AR and VR applications using software development tools and libraries. - identify and mitigate the potential risks and challenges associated with AR and VR, such as VR sickness and privacy concerns. - design and evaluate AR and VR experiences that are both immersive and engaging for users.
Text Books - Chetankumar G shetty, Augmented Reality, McGrawHill, 2020 - Steven M. Lavalley, Virtual Reality, Cambridge University Press, 2016 - Steve Marschner and Peter Shirley, Fundamentals of computer graphics, CRC Press, 2015
Reference Books & Websites https://learn.unity.com/pathways

Evaluation Scheme:

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Sr. No.	Evaluation Elements	Weightage (%)
1	MST	25
2	EST	40
	Evaluation Elements	Weightage (%)

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